IT2226 AI Innovation & Deep Learning Project

ShopSphere Project Plan

Win-Back Decision Agent

From problem to prototype, organised by Design Thinking phase. Built from the ShopSphere brief, the IT2226 Project Guidebook, the Learning Plan, the data pack and policy documents, and the Design Thinking worked example.

1. Empathise	2. Define	3. Ideate	4. Prototype	5. Test

0	**Project essentials**	Dates, marks, rubric, questions for Mr Wee
1	**Empathise**	Persona, stakeholders, Empathy Synthesis Sheet
2	**Define**	Contradictions, needs, POV, How-Might-We, success measures
3	**Ideate**	Crazy 8s, impact/feasibility, ranked shortlist
4	**Prototype**	The model, the AI agent, the app and dashboard
5	**Test**	How we test the model and the agent; pass/fail criteria
A	**Appendix A**	Problems ranked, start order, traps
B	**Appendix B**	Policy rules quick reference

How to read this document

- Each Design Thinking phase starts with a coloured banner. The tracker under it shows where you are.
- Each phase ends with its **Deliverable** box. That box is what the next phase starts from.
- Quotes marked SAYS are verbatim from the brief. THINKS and FEELS are our inferences, to be validated.

SECTION 0

Project essentials

Read this first: dates, marks and what is assessed

The documents disagree on dates

Milestone	Guidebook + Learning Plan (carry the marks)	ShopSphere brief
Model review	Progress Review 1, Week 5 (w/c 16 Nov 2026)	Model checkpoint, Week 8
Prototype	Progress Review 2, Week 14 (w/c 18 Jan 2027)	Live demo, Week 13
Final	Week 18 in the schedule, Week 19 in the rubric (15 Feb 2027)	Week 16

Rule: plan to the earlier date every time and treat the later one as buffer. Confirm with Mr Wee. The brief also says both "17 weeks" and "16 weeks".

Timeline and what to have ready

Week	Milestone	Weight	What to have ready
2	Set-up	—	Team of 4 and a leader; repo with every member committing from their own account; each member's AI Use Log; task board; weekly meeting log
3	Idea presentation	—	Proposal (max 6 pages) and 10-min pitch; individual technical spec; group slides
5	**Progress Review 1**: supervised model	**20%** (15 ind. + 5 group)	EDA and data prep; custom-trained model with tuning; evaluation against a baseline; notebook; live walkthrough and task
7	Mock interview	**10%**	Date to confirm (Learning Plan places it around 30 Nov)
8	Model checkpoint (brief)	—	Short report and notebook; first test-set submission; chosen model with justification
13	Prototype demo (brief)	—	Treat as the rehearsal for PR2
14	**Progress Review 2**: agent and app	**30%** (25 + 5)	End-to-end app; remotely deployed model; dashboard; early evaluation of model and agent
18/19	**Final**	**40%** (35 + 5)	Report (max 25 pages); 20-min demo where every member presents; reproducible repo; final AI Use Log; Ethical AI Checklist; reflection; peer review

Where the marks sit

Individual work is **75%**, group work **15%**, the mock interview **10%**. About **30%** can only be earned live, with no notes or AI: a code walkthrough, a tutor-set change, a seeded-fault debug and questions on your AI Use Log. A polished app you can't explain earns little.

Checkpoint	Individual criteria	Group
PR1 (Week 5)	Input Data 5 · Supervised Model 5 · Source Codes 5	Progress, slides, delivery 5
PR2 (Week 14)	Supervised Model 5 (A band: ≥80% accuracy *or industry benchmark*, deployed **remotely**) · Gen AI / agent 5 · **Application 10** · Source Codes 5	Slides, delivery 5
Final	Presentation 5 · App Demo 10 · Ethical AI 5 (A band: ≥90% relevant checklist answers) · Reflection 10 · Peer Review 5	Slides, delivery 5

Mark levers

- Write the AI Use Log "Why" column as you go: a specific check, finding and action. "Accepted, it worked" scores nothing.
- A log that contradicts your commit history counts as misconduct.
- Commit small and often from your own account. Commit bulk AI-generated code separately from your own edits.
- Run weekly mock walkthroughs: open a random file and explain what it does, what depends on it, what breaks without it.
- Keep a running reflection log from now. Reflection is worth 10 marks and is hard to write at the end.

Confirm with Mr Wee (wee_chee_hong@nyp.edu.sg)

1. Which milestone dates apply, and the exact PR1 submission time.
2. How individual marking works with one team solution. Safest split: every member owns a DL model variant plus an agent tool, on a shared data pipeline.
3. When the mock interview is.
4. Whether the guidebook's "Gen AI capability" means the brief's agent (we assume yes).
5. Whether PR-AUC/F1 can be the "industry benchmark". Only 14.6% of customers buy, so predicting "no" for everyone already scores 85.4% accuracy.
6. Which LLM APIs are approved, the API budget, and what counts as remote deployment.
7. The "Individual Contribution Evidence" criterion (Guidebook §7.4), which is not in the rubric table.

PHASE 1

Empathise

Who has the problem, and what do they say, think, do and feel?

1. Empathise	2. Define	3. Ideate	4. Prototype	5. Test

Starting scenario

Stakeholder	ShopSphere staff, mainly the CRM and Retention team
Constraints	A DL model plus an agent that uses it; no real messaging or payment connection; humans approve; the data pack is not altered; open-source tools only
Success measure	Staff make weekly retention decisions that are timely, policy-compliant, cost-effective and provably working

Primary persona: "Maya", the CRM team

A composite of the CRM Manager and two CRM Executives who build the Monday win-back list.

SAYS

"By the time the list is ready, it's already out of date."
"We keep giving vouchers to people who would have bought anyway."
"Last quarter we sent a 20% voucher to Bronze customers by mistake. The policy says Platinum only. Nobody noticed for weeks."

THINKS

"Days since last order is the best signal I've got."
"If I misread the policy, it's on me."
"A flat 10% is safe, even if wasteful."

DOES

Exports from four systems into spreadsheets, matches them by hand, picks customers by time since last order, and sends most of them the same 10% voucher. Takes about 1.5 days.

FEELS

Pressure to finish every Monday; anxiety about silent policy breaches; frustration that the effort produces a stale list with no proof it works.

Secondary stakeholders

Who	What they say	What they do / the gap
Finance Controller	Wants to know "before the money is spent" that it goes to the right customers, and "afterwards whether it worked"	Sets weekly per-country budgets with no pre-check and no after-the-fact measure
Customer Service Lead (~30 agents)	A lost-parcel customer got a "We miss you, here's 10% off" email the same day they complained	Agents look up orders, deliveries and tickets in three systems; volumes double in Nov/Dec
Head of Marketing	"I find out a month later" what's trending	Builds a monthly deck from exports; can't compare channels, regions or seasons
Operations Manager	"I suspect some carriers are costing us customers... I can't prove it"	Late deliveries are tracked but not linked to customer behaviour
Merchandising Lead	Marketing promoted a zero-stock product and discounted thin-margin items	Stock and margin never reach campaign planning
The customer (indirect)	Not quoted	Receives the contact. A complaint followed by a win-back offer shows the cost of getting it wrong

DELIVERABLE 1 · Empathy Synthesis Sheet

Verbatim quotes: the CRM, Customer Service and Finance quotes above.

Observed behaviours: manual exports from four systems, recency-only selection, flat vouchers, a monthly marketing deck, delivery data not linked to customers.

Aha contradiction: staff know what good looks like, but the information and the rules aren't in front of whoever decides, at the moment they decide.

Hands over to Define: the quotes and the aha become the evidence behind the problem statement.

PHASE 2

Define

Turn the evidence into a point of view and a How-Might-We question

1. Empathise	**2. Define**	3. Ideate	4. Prototype	5. Test

The contradictions

What people say or intend	What actually happens	The insight
CRM wants to target customers who need a nudge	Selection is by recency, with a flat voucher	The rule can't tell at-risk customers from would-buy-anyway ones. The problem is the missing signal, not effort.
"The policy says Platinum only"	A 20% voucher went to Bronze and nobody noticed for weeks	Policy lives in documents and people's heads, not at the moment of decision, and failures are silent.
Retention wants customers back	A "we miss you" email landed on a customer with a fresh complaint	Hand-matching lets the customer's current state slip past whoever chooses who to contact.
Marketing spend keeps rising	Results arrive in a monthly deck	Money is committed daily but judged monthly, so decisions are made blind.
Ops suspects carriers lose customers	Deliveries are tracked but not linked to behaviour	The evidence exists but hasn't been joined.

Overall aha

ShopSphere has the data and the policies, but they sit in separate systems. Decisions rest on one signal (recency), with no check on whether a nudge is **needed**, **allowed** or **worthwhile**, and no feedback on whether it worked. The three gaps: **foresight**, **compliance at the point of decision**, and **feedback**.

Needs extracted

#	Need	Wish
N1	Know early who is likely to buy or drift, and why, in terms staff can act on	W1, W2
N2	Spend each country's weekly budget where it changes behaviour	W3
N3	Trust that nothing breaks policy, and nothing goes out unapproved or without a recorded reason	W4, W9
N4	See the whole customer state before acting	W5
N5	See what's working now, not a month later	W6, W10
N6	Get cross-system answers without hand-matching	W7

POV statement

User	ShopSphere's CRM team
Need	needs a way to decide each week which customers to re-engage, with what offer and channel, based on who is actually likely to buy or drift, and checked against policy and budget before anything goes out,
Insight	because today they pick mainly by days since last order from hand-matched spreadsheets, so lists are stale, vouchers go to people who'd have bought anyway, and policy breaches go unnoticed.

How-Might-We candidates

Tags map each HMW to the brief's problem statement: [A] decisions made without knowing who will buy · [B] information scattered and matched by hand · [C] no timely view of what's working · [D] decisions must be timely, consistent, compliant and cost-effective.

#	Focus	How might we…	Wish
1	Whole loop [A, D]	…help the CRM team decide each week who to re-engage, how and at what cost, so every decision is timely, compliant, within budget and explainable?	W1–W4
2	Foresight [A]	…give staff an early, trustworthy sense of who is about to buy and who is slipping away?	W1, W2
3	Cost-effectiveness [A, D]	…make sure retention money reaches customers it can actually influence, not those who'd buy anyway?	W3, W10
4	Policy at decision [B, D]	…make sure every decision follows policy when it's made, not weeks later?	W4
5	Customer's situation [B]	…make sure a win-back message never lands on someone who just complained or opted out?	W4
6	Timeliness [B, D]	…get each week's decisions made in time to matter, the same way every week?	W3
7	Seeing what works [C]	…let managers see, while there's time to change course, whether decisions are working?	W6, W10
8	Trust and control [D]	…make recommendations staff understand and trust, while keeping people in charge?	W2, W9

Supporting HMWs: tell a customer who is quietly drifting from one who buys in bursts · match each customer to the smallest nudge that works, including no nudge · check consent, frequency, complaints, tier, stock and margin every time without relying on memory · compare channels, regions and seasons in one place · show delivery performance next to customer behaviour · record why each decision was made.

Validate with the data (hypotheses)

#	Hypothesis	Tables
1	Recency predicts October purchase (this is also the baseline)	transactions, train
2	Past contacts landed within days of an open ticket	support_tickets, retention_contact_history
3	Past offers broke tier eligibility (**data cleaning found ~134 cases**)	offer_catalogue, customers
4	Contacts went to do-not-contact or opted-out customers	customer_preferences
5	Late or lost deliveries lower next-30-day purchase, by carrier	deliveries
6	Past voucher recipients buy anyway (beware selection bias)	retention_contact_history
7	"One in three inactive for 90 days" is true	sessions

DELIVERABLE 2 · Problem Statement Card

SELECTED HOW-MIGHT-WE (merges HMW 3 and 4)

"*How might we make sure each week's retention budget reaches the customers it can actually influence, with an offer that policy allows, before anything is sent?*"

Folded into the concept: HMW 2 (foresight) becomes the prediction model; HMW 5 (customer's situation) becomes the policy gate; HMW 8 (trust and control) becomes the approval step.

Parked: HMW 7, because the W6 dashboard covers it · HMW 6, because automating the list solves it as a side effect.

Success measures (locked before building): the DL model beats the non-DL baseline on held-out data · zero policy violations across test scenarios · no weekly list exceeds a country's budget · every recommendation is approved and has a recorded reason · list preparation drops from ~1.5 days to minutes.

PHASE 3

Ideate

Generate options against the HMW, score them, and pick one

1. Empathise	2. Define	**3. Ideate**	4. Prototype	5. Test

Constraints we filter ideas against (from the brief)

- There must be a deep learning model, and an agent that uses it.
- No real messaging or payment systems. A human approves every decision.
- The data pack is not altered. At least 5 wish-list items, including W1, W3 and W6.

Crazy 8s against the HMW

1 Win-back decision agent Predicts who will buy, decides on a voucher, checks policy	**2 RFM scorecard** Rank by recency, frequency and spend. No ML
3 Uplift model Predict who buys *because of* a voucher	**4 "Ask ShopSphere" chatbot** Answers staff questions across systems (W7)
5 Customer 360 panel One-screen summary for service agents (W5)	**6 Auto-campaign bot** An LLM writes the messages and sends them
7 Service-failure trigger A late or lost delivery gets a service-recovery credit, not "we miss you"	**8 Budget optimiser** Spend each country's budget where the expected return is best

Method: 8 ideas in 8 minutes, then dot-voting on an Impact / Feasibility grid.

Impact / feasibility scoring

Idea	Impact	Feasibility	Result
1. Decision agent	High	High	**Core concept**
8. Budget optimiser	High	High	Merged into 1
7. Service-failure trigger	Medium	High	Merged into 1
3. Uplift model	High	Low: the data has no control group	Runner-up, parked as future work
4. Chatbot	Medium	Medium	Parked as a stretch goal (W7)
5. Customer 360	Medium	High	Parked (W5)
2. RFM scorecard	Low	High	✗ Breaks a constraint (no DL). Kept as the baseline
6. Auto-campaign bot	Medium	Medium	✗ Breaks a constraint: sends real messages without approval

Traceability: Define → Ideate

Evidence from Empathise	Feature in the concept
"We keep giving vouchers to people who would have bought anyway."	Probability bands decide whether an offer is needed at all
"We sent a 20% voucher to Bronze by mistake… nobody noticed for weeks."	Tier caps checked in code, before the list is shown
Lost-parcel customer got a 10%-off email the same day	Open-ticket block, plus a service-recovery offer
"I want to know before the money is spent…"	Budget fit and the approval step
"By the time the list is ready, it's already out of date."	The list is built in minutes, every Monday

DELIVERABLE 3 · Ranked Concept Shortlist

Selected: the Win-Back Decision Agent (ideas 1, 7 and 8 combined). Every Monday it turns model predictions and company rules into a re-engagement list that fits the budget. A CRM Executive approves it before anything is sent.

Why it beat the uplift model: an uplift model needs customers who were deliberately *not* contacted, and ShopSphere's contact history has none. The decision agent works with the data that exists, tackles voucher waste with probability bands, and fits every constraint in the brief.

Hands over to Prototype: the selected concept and its riskiest assumption.

PHASE 4

Prototype

What we will build: the model, the agent, the app

4.1 Overview

A tool for ShopSphere's CRM team. Every Monday it produces a recommended re-engagement list for each country, saying for each customer: whether to contact them, which offer (if any), which channel and language, and why. The list stays within each country's weekly budget, follows every policy, and goes nowhere until a CRM Executive approves it.

Part	What it is	Wish items
A. Prediction model	Supervised deep learning. Predicts who will buy in the next 30 days, and why	W1, W2
B. AI agent	Uses the model and the company rules to build the list, explain decisions and answer staff questions	W3, W4, W9
C. App and dashboard	Staff review and approve the list, and see trends	W6
Stretch goals	Draft messages · customer summary · plain-language Q&A	W8, W5, W7

Hard constraints (never break these)

- The main model is deep learning, compared against at least one non-DL baseline. The agent uses the trained model.
- Nothing is ever sent: no connection to real email, SMS, WhatsApp or payment systems.
- A human approves every list, and every decision has a recorded reason.
- The data pack is never altered; any synthetic data is labelled. Open-source libraries and approved APIs only; no API keys in the repo.
- Only data dated on or before **30 Sep 2024** is used for model features.

4.2 Part A: the purchase prediction model

Type	Supervised binary classifier
Input	One row of features per customer, built from data up to 30 Sep 2024
Output	Probability the customer buys at least once between 1 and 30 Oct 2024
Labels	train.csv: 64,000 labelled customers (target 1 = bought). test.csv: 16,000 to predict
Class balance	14.6% buy, 85.4% don't
Main model (DL)	A multilayer perceptron (MLP) on the feature table, as the brief requires
Baselines (ML)	Recency rule (what CRM does today) · logistic regression · LightGBM or XGBoost
Member variants	Plain MLP · MLP with embeddings for categories · GRU/LSTM over weekly activity · wide-and-deep or TabNet

The model only predicts. It gives each customer's probability and the reasons, and ranks customers. It never decides who gets a voucher, checks policy, chooses offers or checks the budget. The agent and rule functions do that.

Goals

1. Beat every baseline on held-out data. Main metric PR-AUC, with F1, precision, recall and ROC-AUC.

2. Produce **calibrated** probabilities: if it says 0.7, about 70% of those customers buy. The voucher bands depend on this.
3. Explain each prediction in terms staff can act on (W2), e.g. "no sessions in 60 days", "2 late deliveries".
4. Perform fairly across countries and tiers (Ethical AI Q9).
5. Work as a tool for the agent: saved, loadable, and able to score all 80,000 customers in seconds.

Data contract: features.csv

- One row per customer for all 80,000 customers. Key: **customer_id**. The target is not in the file; the notebook joins train.csv.
- Numbers are numeric, flags are 0/1, categories are text. No raw dates: convert them first (e.g. days_since_last_order).
- Keep **average order value**: the agent's cost formula needs it.
- Built in KNIME: each event table is summarised per customer with GroupBy, then left-joined onto the cleaned customers table. Customers with no activity get 0 for counts and a flag where something never happened (e.g. never_ordered).

How to build it

1. Load features.csv and join train.csv.
2. Split **70/15/15** into train / validation / test, stratified on target, fixed seed. Every member uses the same split.
3. Preprocess with a scikit-learn Pipeline: median imputation, StandardScaler on numbers, one-hot encoding. Fit on train only.
4. Train the baselines: recency rule, logistic regression (class_weight="balanced"), LightGBM.
5. Train the MLP: Dense 128 → Dropout 0.3 → Dense 64 → Dropout 0.3 → sigmoid. Weighted binary cross-entropy (pos_weight ≈ 5.85), Adam 1e-3, batch ~512, early stopping on validation PR-AUC (patience 5).
6. Tune learning rate, layers and width, dropout, weight decay, batch size. Log every run: this is the evidence of optimisation.
7. Calibrate the probabilities (Platt or isotonic, fitted on validation). Class weighting distorts probabilities.
8. Set the decision threshold and band cut-offs on validation; show the actual purchase rate rises band to band.
9. Evaluate once on the test split (see Phase 5).
10. Explain: permutation importance overall, SHAP for each customer's top reasons.
11. Save the pipeline and model. Write **predict(customer_ids) → probability, band, top_reasons**, the tool the agent calls. For PR2, put it behind a FastAPI endpoint and deploy remotely.
12. Optionally, write test.csv predictions in the sample_submission format.

Leakage rules

- 510 delivered_date values fall after the cutoff: they are blanked, and the original delivery_status is never a feature.
- Planned campaigns starting after the cutoff are excluded from features.
- Imputation and scaling are fitted on training rows only. Nothing is tuned on the test split.

4.3 Part B: the AI agent

Every week, for each country, the agent scores customers with the model, removes anyone policy says must not be contacted, decides whether each remaining customer needs an offer, picks the smallest suitable offer their tier allows, costs it, fits the list into the weekly budget, writes a reason for every decision, and hands the list to a CRM Executive. Staff can also ask it questions, e.g. "Why is customer X excluded?" or "Rebuild DE's list with 10% less budget."

Goals

Zero policy violations · no country over budget · no vouchers for customers who'd buy anyway · a reason for every recommendation · nothing sent without approval · the list built in minutes, not 1.5 days.

> **Key design principle**
> **The LLM orchestrates and explains. It never makes policy decisions.** Policy rules, offer eligibility, costs and budget checks are plain, deterministic Python functions. The LLM calls them and cannot override their results. This guarantees compliance, and is a strong point for the responsible AI section.

Stack: LangGraph (single agent with tools; multi-agent optional) · lecturer-approved LLM API, key in a .env file never committed · Pydantic to validate inputs and outputs · optionally MCP to expose the tools.

Agent tools

Tool	Returns
predict_purchase(customer_ids)	Probability, band and top reasons, from the model
get_customer_profile(customer_id)	Tier, country, preferences, average order value, recent orders, deliveries, tickets and offers
check_policy(customer_id, offer_code, channel, week_start)	Allowed (true/false) and a list of reasons
select_offer(customer_id, band)	An offer code, or none, with the rationale
estimate_offer_cost(customer_id, offer_code)	Expected cost, using the policy formula
allocate_budget(country, week_start, candidates)	The final list, within budget
build_weekly_list(week_start, country)	Runs the tools above in order; returns the draft list
record_decision(list_id, customer_id, action, approver, reason)	Writes to the audit log
draft_message(customer_id, offer_code) *(stretch, W8)*	Text in the preferred language and channel, for review only. Never sends

Decision logic for each customer

Step	What happens
1. Predict	Get probability, band and reasons from the model.
2. Policy gate	Exclude (reason recorded) if: do_not_contact = Y · channel not opted in · open High-priority ticket (route to Customer Service) · an offer counting toward the monthly limit in the last 30 days · already 2 messages this week.
3. Is an offer needed?	Service failure in last 60 days → OFF-SRC ($10 credit). **High** band (would buy anyway) → reminder only (OFF-NONE) or skip. **Middle** band (on the fence) → smallest offer likely to work: OFF-FS or OFF-V10. **Low** band (drifting) → stronger offer only if high-value; otherwise reminder or skip. Cut-offs set on validation (0.6 / 0.2 are placeholders).
4. Tier cap	Bronze max 10% · Silver and Gold max 15% · Platinum max 20% (full table in Appendix B).
5. Product rules	No vouchers on non-promotable items; don't feature out-of-stock or low-stock items.
6. Expected cost	Policy formula (Appendix B). Redemption rate estimated from contact history per offer and tier; 10% if it can't be estimated. Any offer above $25 is flagged for the CRM Manager.
7. Fit the budget	Rank by value for money: service recovery first, then middle band, then high-value low band; within each, by average order value. Add customers until the weekly budget runs out. No carry-over.
8. Channel and language	Preferred channel if opted in, else another opted-in channel, else exclude. Preferred language. No SMS/WhatsApp between 9 pm and 8 am local time. Unsubscribe option; no false urgency.
9. Human approval	The CRM Executive sees offer, channel, language, cost and reason per customer; approves, edits or removes. Everything is logged with approver and timestamp.

Guardrails

- No function exists that sends anything. The LLM cannot change the result of check_policy or allocate_budget.
- All tool inputs and outputs are validated with Pydantic. Tool-call and token limits per run stop loops and runaway cost.
- Out-of-scope requests are refused. Every tool call and decision is logged.

4.4 Part C: the app and dashboard

- **App:** Streamlit (simplest), or a FastAPI backend with a separate frontend.
- **Pages:** weekly list review and approval · customer detail with prediction reasons and history · ask the agent · dashboard.
- **Dashboard (W6):** sales, traffic, campaign performance, delivery performance and customer activity, by country and over time; delivery performance by carrier; prediction bands by country.
- **Deployment:** Docker, deployed remotely (e.g. Hugging Face Spaces or Render) for PR2. Remote deployment is needed for the A band.

4.5 What we need

Need	Detail
Data	Cleaned KNIME tables, features.csv, train/test, offer_catalogue, retention_budget, retention_contact_history, customer_preferences, support_tickets, deliveries, products, the 4 policy documents
Compute	Google Colab or Kaggle Notebooks for the model
Libraries	pandas, scikit-learn, PyTorch or Keras, LightGBM, SHAP, matplotlib/plotly, Optuna (optional) · LangGraph, Pydantic · FastAPI, Streamlit, Docker
Access	A lecturer-approved LLM API key and budget
Team agreements	Feature table format · shared split and seed · metrics · who owns which model variant and agent tool · repo structure
Evidence	Notebook with markdown · tuning log · comparison table · scenario test results · each member's AI Use Log · small commits from own accounts

4.6 Build order

When	What
PR1 (Week 5)	Cleaned data and feature table · baselines · each member's DL model · evaluation · progress slides
Before PR2	Calibration and bands → predict() → policy, offer, cost and budget functions with unit tests → agent in LangGraph → app and dashboard → scenario test suite
PR2 (Week 14)	Agent and app working end to end · model deployed remotely · evaluation results
Final	Report · live demo · Ethical AI checklist · reflection · reproducible repo with README

4.7 Limitations (state openly)

- The model predicts **buying**, not **responding to a voucher**. The bands are a heuristic, not proof an offer caused a purchase.
- It's trained on October behaviour, but the budget covers October to December, including the Nov/Dec peak.
- Past offers weren't sent at random, so any analysis of how customers responded is biased.
- The "2 messages per week" rule includes campaign emails, which the data covers only partly, so we approximate it.

DELIVERABLE 4 · Prototype Package

Riskiest assumption being tested: customers with a high predicted probability buy without a voucher, so giving them one wastes money.

What we test with: the model on the held-out test split, and the agent on a scenario suite of realistic ShopSphere cases (Phase 5).

Hands over to Test: the pass/fail criteria below, set before testing.

PHASE 5

Test

How we prove the model and the agent work, with criteria set in advance

5.1 Testing the model

Keep a test split the model never sees

- **Train** (70%): the model learns. **Validation** (15%): tuning, early stopping, threshold, calibration. **Test** (15%): used **once**, at the end, for the reported numbers.
- test.csv has no labels, so it can't be scored locally. Ask whether there is a leaderboard.
- Optional: stratified 5-fold cross-validation, reported as mean ± standard deviation, to show the score is stable.

Compare against baselines

Model	What it shows
Always predict 0	The floor: 85.4% accuracy and PR-AUC ≈ 0.146. This is why accuracy alone means nothing
Recency rule	What the CRM team does today. Beating it is the business case
Logistic regression	The simple ML baseline (required)
LightGBM	A strong ML baseline. If it beats the MLP, say so and explain why
MLP and variants	The DL model

Metrics and how to read them

Metric	What it tells you	Random scores
PR-AUC (main)	How well it finds buyers when buyers are rare	0.146
ROC-AUC	How well it ranks buyers above non-buyers	0.5
Precision / recall	"Of those we flag, X% buy" / "We catch X% of all buyers"	
F1, confusion matrix	Balance of precision and recall at the chosen threshold; counts of each error	
Precision in top 10% / lift	Quality at the top of the list, which is all the budget can cover	
Accuracy	Report only next to the 85.4% floor	

Translate results into business terms, e.g. "Of the top 1,000 customers we flag, 620 buy, versus 146 at random."

Checks that make the numbers trustworthy

Check	How	Pass looks like
Calibration	Reliability diagram; Brier score before and after calibration; purchase rate per band	Points near the diagonal; rate rises low → middle → high
Overfitting	Train vs validation loss and PR-AUC per epoch	Validation doesn't get worse while train keeps improving
Stability	Train the MLP with 3–5 seeds	Small standard deviation
Leakage	Suspiciously high scores; one dominant feature; shuffled-label test; scaler/imputer fitted on train only	ROC-AUC not above ~0.95 without explanation; shuffled-label PR-AUC ≈ 0.146
Fairness (Ethical Q9)	Recall, precision, PR-AUC by country, tier and age group (incl. unknown and age 18)	No large gaps, or gaps explained with a mitigation
Error analysis	Inspect false negatives and false positives	Patterns identified (e.g. new customers with little history)
Ablation	Retrain without one table's features at a time	Shows which data sources matter (supports "≥4 sources")
Explanations	Sanity-check SHAP reasons	Directions make sense (e.g. recent activity raises probability)

What to report

- One comparison table: every model × every metric, on the test split.
- Plots: PR curve (all models), confusion matrix, reliability diagram, learning curves, feature importance, metrics by country.
- The tuning log, and one slide on leakage checks and their results.
- **For PR2:** unit-test predict(): same output as the notebook, handles an unknown customer ID, returns values between 0 and 1, scores 80k customers fast enough.

5.2 Testing the agent

Scenario	Expected result
Bronze customer proposed for OFF-V20	Blocked
Customer with an open lost-parcel ticket	Excluded and routed to Customer Service
do_not_contact customer	Excluded
Customer contacted 10 days ago	Excluded, unless the offer is OFF-NONE
Prefers SMS but SMS opt-in = N	Switched to another opted-in channel, or excluded
High-probability customer	No voucher
Candidates exceed the weekly budget	List trimmed to stay within budget
Offer with expected cost above $25	Flagged for the CRM Manager

Metrics: policy violation rate (target 0) · budget adherence (100%) · decisions with a recorded reason (100%) · tool-call accuracy · task success rate · time to build a list · cost per run · explanation faithfulness (does the written reason match what the tools returned?).

5.3 Testing the riskiest assumption

In retention_contact_history, compare purchased_within_14d for voucher contacts versus reminder-only contacts, within each predicted-probability band. Note the selection bias: past offers weren't sent at random.

PASS if (set in advance)

- The DL model beats every baseline on PR-AUC on the held-out test split, with no leakage found.
- 0 policy violations across all scenarios, including the Bronze-20% and open-complaint cases from the brief.
- No country goes over its weekly budget.
- 100% of recommendations have a recorded reason and need approval.
- The list is built in minutes, not 1.5 days.

FAIL if: the DL model does not beat the baselines, any policy violation reaches an approved list, or any budget is exceeded.

DELIVERABLE 5 · Verdict → Solution Brief (complete after testing)

Section	Content
Scenario	CRM builds a stale, recency-based win-back list by hand; vouchers are wasted and policy breaches go unnoticed
POV / Problem	Data and rules sit in separate systems, so decisions rest on recency alone with no check on need, policy or worth
Solution	Win-Back Decision Agent: DL purchase model + rule-checked offer, cost and budget tools + human approval
Evidence	[model results vs baselines] · [scenario test results] · [time to build a list]
Verdict	[Validated / Partly validated / Not validated]
Next step	[e.g. a randomised hold-out group so an uplift model becomes possible]

APPENDIX A

Problems ranked, start order and traps

Planning reference

#	Problem	Why it ranks here
1	Data foundation: clean, join and build leakage-safe features (nothing after 30 Sep 2024)	Everything depends on it. PR1 Input Data (5 marks); leakage is an A-band catch; requirement 4 needs ≥4 sources
2	Predict purchase in the next 30 days (W1) and explain why (W2), DL vs a non-DL baseline	Mandatory; earns PR1 and PR2 marks; the agent needs it as a tool
3	Weekly who / which offer / which channel within budget (W3), with approval and reasons (W9)	Mandatory; sits in the largest-mark area at PR2 (Application 10 + agent 5). A prediction alone isn't a decision
4	Policy compliance (W4): consent, frequency, complaints, tier, stock and margin	Rules live in 4 policy documents plus tables. Two incidents in the brief are ready-made test cases
5	Dashboard and a prototype staff could really use (W6)	Mandatory; needs no model, so it can start in parallel; drives UX marks
6	Evidence: evaluating model and agent; campaign impact (W10) is a stretch	Weak evidence loses marks in the report and live defence
7	Responsible AI: fairness, privacy and consent, transparency, risk of being wrong	5 marks at the Final plus a report section; log decisions from now
8	Scoping the wish list: ≥5 items including W1, W3, W6	W2, W4 and W9 come almost free from problems 2–4

Start order

1. Fix scope and finish Empathise and Define.
2. Audit every table: types, true/false encodings, numbers stored as text, duplicates, missing values, outliers, join keys, noise_* columns.
3. Build the one-row-per-customer feature table with a strict 30 Sep cutoff, and fix the split.
4. Train a non-DL baseline first, then the DL model.
5. In parallel: one member extracts the policy rules; another starts the dashboard data.
6. Build the agent on top of the model and the rules.

Traps

- customers.csv has pre-aggregated columns (loyalty_score, email_open_rate, discount_usage_rate, avg_review_score). Check they don't use post-cutoff information.
- train.csv is labelled and test.csv isn't. Never tune on test.
- Don't judge the model on accuracy alone: 85.4% of customers are 0.
- The model is trained on October behaviour, but the budget plan covers Q4, including the Nov/Dec peak.
- Likely to purchase is not the same as worth a voucher. The probability bands in the concept address this.

APPENDIX B

Policy rules quick reference

From the 4 policy documents, offer_catalogue and retention_budget_2024Q4

Never contact

- do_not_contact = Y · email offers to email_opt_in = N · SMS offers to sms_opt_in = N.
- Customers with an **open High-priority ticket** (lost parcel or payment issue): Customer Service handles them first.
- Customers sent a retention offer in the last **30 days** (offers with counts_toward_monthly_limit = 0, i.e. OFF-NONE, are exempt).

Offers and tier eligibility

Code	Offer	Type	Eligible tiers	Expected cost
OFF-NONE	Personalised reminder only	None	All	$0
OFF-FS	Free express shipping on next order	Shipping	All	$6 × redemption rate
OFF-V10	10% voucher, min spend $40, 14 days	Voucher	All	10% × AOV × redemption rate
OFF-V15	15% voucher, min spend $60, 14 days	Voucher	Silver, Gold, Platinum	15% × AOV × redemption rate
OFF-PTS2	Double loyalty points for 30 days	Points	Gold, Platinum	$3 × redemption rate
OFF-V20	20% voucher, min spend $100, 14 days	Voucher	Platinum	20% × AOV × redemption rate
OFF-SRC	$10 service-recovery credit	Credit	All, after a service failure in the last 60 days	$10, full cost

AOV = the customer's average order value. Redemption rate is estimated from retention_contact_history; use 10% if it can't be estimated for a segment. Any single offer with expected cost above $25 needs CRM Manager approval.

Product and stock

- No vouchers on products with is_promotable = No (margin below 18%).
- Don't feature a product with stock_on_hand = 0, or with stock below 50 and the next restock more than 14 days away.

Weekly retention budget (13 weeks, 7 Oct – 30 Dec 2024, no carry-over)

Country	US	UK	DE	FR	IN
Weekly range	$12,000 – 21,600	$6,000 – 10,800	$6,000 – 10,800	$3,000 – 5,400	$2,500 – 4,500

Contact rules

- Use preferred_channel and preferred_language where consent allows.
- At most 2 marketing messages per customer per week, including campaign emails.
- No SMS or WhatsApp between 9 pm and 8 am customer local time. Every message has an unsubscribe option and no false urgency.
- Every list is approved by a CRM Executive; automated lists record each customer's selection reason and offer.

Related service and shipping rules

- A customer with 2+ tickets in 30 days is "at risk"; agents must check their history before replying.

- Platinum customers with an open High-priority ticket are escalated straight away.
- Customer Service must be able to see any offer sent in the last 14 days.
- 2+ late deliveries in 60 days qualifies the customer for OFF-SRC without complaining. A lost parcel gets a refund or replacement plus OFF-SRC.
- Loyalty tiers by 12-month spend: Bronze under $120 · Silver $120–399 · Gold $400–899 · Platinum $900+.